Certifying rare overflow risk of regional pickup-and-delivery plans

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Draft, October 2026

Abstract

Medical, cold-chain and dangerous-goods distribution require that a vehicle routing plan meets service levels of $1-\gamma$ with γ between 10^{-3} and 10^{-6}. We study how to certify such levels for a given plan of the vehicle routing problem with simultaneous pickup and delivery, where each route overflows if its peak load exceeds capacity. Under gamma demands with a Gaussian factor copula we show that, on the logarithmic scale, a route overflows in one of two ways — at the start, through its total delivery, or at the end, through its total pickup — with explicit scales given by secular equations, and that middle stages and within-stage combinations are close behind on long routes. This structure explains a silent failure of single-shift importance sampling: when the weaker end scale exceeds a quarter of the stronger one the estimator loses efficiency, and at practical levels it underestimates by the mass of the missed mode while reporting a small standard error. We propose a closed-form stage-mixture estimator that places one component at the exact dominating point of each relevant failure mode, needs no pilot simulation, and extends to a route-day simulator with stochastic travel and service times and time windows, where no exact method exists, and to whole hubs whose routes share demand factors. On 148 instances built from population, postal and road data for Northern Vietnam and Guangdong (151 million people), the estimator matches exact dynamic programming and 10^7-draw Monte Carlo within statistical error, has relative variance 4–6 at every level, and costs two to three orders of magnitude fewer samples than cross-entropy mixtures at 10% relative precision. We also report that quantum amplitude estimation, often proposed for such tasks, does not beat this classical estimator on these instances.

1 Introduction

Planning models for stochastic vehicle routing control risk through chance constraints, robust counterparts or distributionally robust ambiguity sets [Gounaris et al., 2013, Dinh et al., 2018, Ghosal and Wiesemann, 2020]. They evaluate route risk inside the optimiser, at moderate levels and with exact formulas, approximations or plain sampling. Applications with strict service levels need a different step after optimisation: a certificate that the plan actually produced meets $\mathbb{P}(\text{failure}) \le \gamma$ for $\gamma \in [10^{-6}, 10^{-3}]$, computed for every route and for the plan as a whole, at the scale of a regional network.

At such levels crude Monte Carlo needs of the order of $100/\gamma$ draws per route for 10% relative error, which is prohibitive for thousands of routes. Importance sampling (IS) is the standard remedy, and for Gaussian-driven models mean shifts tuned by the cross-entropy (CE) method are the practitioner's default [de Boer et al., 2005]. The difficulty is that the failure of a pickup-and-delivery route is a union of events — an overflow can occur after any stop — and single-mode IS can then fail without warning [Glasserman and Wang, 1997]. Mixtures over failure modes [Owen

et al., 2019, Geyer et al., 2019] cure this when the modes are known, at the price of pilot simulation when they are not.

Contributions. (i) We formulate the certification of a given plan at service levels 10^{-3}–10^{-6}, route-wise and hub-wise. (ii) We characterise the large deviations of the peak load of a pickup-and-delivery route under gamma marginals and a one-factor Gaussian copula: the two end modes with explicit spectral scales, and the relative mass of every other stage (Section 4). (iii) We give a sharp criterion for the failure of single-shift IS and a diagnostic for its finite-level bias (Section 5). (iv) We propose a pilot-free stage-mixture estimator (SMIS) built from exact dominating points, its extension to a route-day simulator without exact solution (MDIS), and to hubs with shared factors, with risk attribution to routes (Section 6). (v) We evaluate on a new set of 148 region-scale instances (Section 7). The importance-sampling ingredients are known — dominating points [Sadowsky and Bucklew, 1990], multiple-path failure of single shifts [Glasserman and Wang, 1997], mixtures as subsolutions [Dupuis and Wang, 2007], factor-copula two-step schemes [Glasserman and Li, 2005, Bassamboo et al., 2008] — and we do not claim new IS theory. What is new is their exact form for the pickup-and-delivery peak, the resulting diagnostic, and a certification procedure that is cheap enough for regional plans.

2 Related work

Risk in vehicle routing. Chance-constrained and robust models [Gounaris et al., 2013, Dinh et al., 2018, Ghosal and Wiesemann, 2020] handle route risk inside the optimisation with exact formulas, moment bounds or scenario sampling. Hazardous-materials routing measures risk as accident probability times consequence along links [Erkut and Verter, 1998]. We found no work that certifies an already optimised routing plan at rare levels with IS, and none that treats the mid-route peak of the simultaneous pickup-and-delivery load [Dethloff, 2001] as a rare event. The closest IS precedent in logistics is stockout simulation in production–inventory systems [Glasserman and Liu, 1996].

Rare-event simulation. Efficiency notions and the dominating-point theory are surveyed by Asmussen and Glynn [2007]; Glasserman and Wang [1997] show that a single shift is inefficient when the rare set has several ways of occurring; Dupuis and Wang [2007] build efficient schemes from subsolutions, of which mixtures are instances; Owen et al. [2019] bound the variance of a mixture that samples each event of a union conditionally; CE with Gaussian mixtures handles multimodal failure regions [Geyer et al., 2019, Papaioannou et al., 2019]. For Gaussian factor copulas, Glasserman and Li [2005] combine a factor shift with conditional exponential twisting; Bassamboo et al. [2008] treat extremal dependence.

3 Problem

A route visits customers $1, \dots, L$. The vehicle leaves the depot with all deliveries $d_1, \dots, d_L$ and collects pickups $p_1, \dots, p_L$. The load after stop k is $L_k = D_k + P_k$ with $D_k = \sum_{i>k} d_i$ and $P_k = \sum_{i \le k} p_i$ $(k = 0, \dots, L)$, the peak is $F = \max_k L_k$, and the route overflows a capacity c if $F > c$.

Demand model. Each d_i is gamma with shape $\kappa = 1/\mathrm{cv}^2$ and scale $\theta_i^d = \bar{d}_i/\kappa$, likewise p_i; dependence is a one-factor Gaussian copula per demand type: $d_i = G_{d,i}^{-1}(\Phi(Z_{d,i}))$ with $Z_{d,i} = \sqrt{\rho}\, W_d + \sqrt{1-\rho}\, E_{d,i}$, and similarly for pickups with an independent factor W_p. The input $u = (W_d, W_p, E_d, E_p)$ is standard normal. All customers of a hub share the two factors.

Definition 1 (Certificate)**.** Given γ and a relative precision ε, a certificate for a route (or a hub) is an estimate $\hat{p}$ of its failure probability p with $|\hat{p} - p| \le \varepsilon p$ at confidence $1 - \delta$, together with the check $\hat{p}(1 + \varepsilon) \le \gamma$; for a hub, also the attribution of p to its routes.

4 Structure of the overflow event

For an index set S with gamma scales θ_S let $C_S = \rho \mathbf{1}\mathbf{1}^\top + (1 - \rho)I$ and

$$\lambda(S) = \lambda_{\max}\Big(\Theta_S^{1/2} C_S \Theta_S^{1/2}\Big),$$

the largest root of $1 = \rho \sum_{i \in S} \theta_i/(\lambda - (1 - \rho)\theta_i)$. Write $\lambda_k^d = \lambda(\{i > k\})$ for the delivery block of stage k, $\lambda_k^p = \lambda(\{i \le k\})$ for its pickup block, $\lambda_k = \max(\lambda_k^d, \lambda_k^p)$, and $\lambda^\star = \max(\lambda_0^d, \lambda_L^p)$.

Proposition 1. $\max(D_0, P_L) \le F \le D_0 + P_L$.

Theorem 2. *As $c \to \infty$ with the route fixed,* $\log \mathbb{P}(L_k > c) \sim -c/\lambda_k$ *and* $\log \mathbb{P}(F > c) \sim -c/\lambda^\star$.

So, on the logarithmic scale, the route overflows at the start (delivery block) or at the end (pickup block). With $\varrho = \min(\lambda_0^d, \lambda_L^p)/\lambda^\star$, the weaker end mode carries relative mass $\gamma^{(1/\varrho - 1)(1 + o(1))}$ at level γ (Proposition 4 in the appendix). On long routes the scales of stages near the ends are close to $\lambda^\star$, so they also matter at practical levels.

Remark 1 (Pre-asymptotic regime). At $\gamma \in [10^{-9}, 10^{-3}]$ and cv $= 0.3$ the exact dominating-point energies are only 16–35% of their asymptotic values c/λ (the log corrections of the gamma quantile are large). The scales are therefore a qualitative guide; the estimators below use exact dominating points.

5 When a single shift fails

Mean-shift IS with shift τ uses the proposal $N(\tau, I)$ and weight $w(u) = \exp(-\langle u, \tau\rangle + |\tau|^2/2)$.

Theorem 3. *Let $\tau^\star$ be the dominating point of the dominant end mode. (a) If $\varrho > 1/4$, the shift $\tau^\star$ is not logarithmically efficient:* $\log m_2(\tau^\star) - 2 \log \mathbb{P}(F > c) \ge c\,(4/\lambda^\star - 1/\lambda_{\text{other}})(1 + o(1))$. *(b) If $\varrho \le 1/4$ and the dominant end block has its second eigenvalue at most $\lambda^\star/4$, the shift is logarithmically efficient.*

The extra second moment in (a) comes from a region the proposal almost never visits, so the sample variance does not reveal it; in practice the estimate is biased low by about the missed mode mass with a small reported error. Our measurements match: on 120 real routes the single CE shift is accurate (median error 0.1–0.2%) when $\varrho \le 0.5$ and off by 3–6% in median and up to 40% when $\varrho > 0.75$, which is the case of balanced deliveries and pickups (Northern Vietnam). ϱ is computed in closed form from the route data, so the diagnostic costs nothing.

6 Estimators

SMIS. For every stage k and block, compute the exact minimum-norm point u_m of $\{L_k(u) \ge c\}$ (constrained minimisation started on the block's Perron direction, using the exact quantile transform), keep the modes whose energy $E_m = |u_m|^2/2$ is within $\log(1/\eta)$ of the smallest, and sample from $q = a_0 N(0, I) + \sum_m a_m N(u_m, I)$ with $a_m \propto e^{-(E_m - E_{\min})}$ and a constant defensive weight a_0. No pilot simulation is needed. Theorem 5 (appendix) gives conditions for logarithmic efficiency.

MDIS. The same construction applies to any simulator that exposes its failure modes as scalar functions $g_m(u) > 0$. We use it for a route-day model with lognormal travel times (shared congestion factor), service times that grow with the handled load, and time windows; no exact method applies.

Hubs. The routes of a hub share the factors, so the hub failure $\bigcup_r \{F_r > c\}$ is again a union of stage events. We embed the route modes in the hub input, compute the likelihood ratio with sparse inner products (each component touches the two factors and one route's coordinates), and estimate from the same draws the probability that each route fails and that it is the only one failing — an attribution of the hub risk to routes. Hubs of a region have independent factors, so the region risk is $1 - \prod_h (1 - p_h)$.

7 Experiments

7.1 Instances

We built 148 instances, one per depot area: the 25 provinces of Northern Vietnam (Hanoi split into two hubs) and 122 county-level areas of Guangdong. Service points aggregate Kontur population hexagons (15 000 people per point in Vietnam, 4 000 in Guangdong) by weighted k-means; distances are OSRM road distances. Daily demand rates come from official postal statistics: 0.087 parcels per person per day in Vietnam (3.2 billion items in 2024) and, for Guangdong, 0.34 received and 0.91 sent (national and provincial 2024 volumes); 1 kg per parcel and truck capacities of 15 and 30 tonnes are assumptions. The set has 40 099 service points serving 151 million people. Plans are solved with the deterministic and the sample-average chance gates; we certify the latter.

7.2 Routes under the demand model

On 120 routes (stratified by region and length) and $\gamma \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$, the CE mixture used as a classical rival agrees with exact dynamic programming (fine quadrature over both factors, routes with $L \leq 8$) in 178 of 178 cases, maximal deviation 0.74%. SMIS has relative error at most 0.6% (90% quantile) in every ϱ band, relative variance 3.8–5.9 (CE: 3.8–6.0), 2–7 components, and is built in 0.4–0.9 s. Including CE's pilot draws (0.8–1.2×10^5 per route), CE costs 224–276 times more samples than SMIS at $\varepsilon = 0.1$ and 3.2–3.8 times more at $\varepsilon = 0.01$.

7.3 Route days without an exact method

On 60 routes with thresholds that split the failure probability evenly between load and time modes, MDIS deviates from 10^7-draw crude Monte Carlo by 0.76% in median (1.77% at the 90% quantile, within the Monte Carlo error of about 1%) at $\gamma = 10^{-3}$; CE deviates by 0.89% and 1.75%; the single CE shift by 2.2% and 22%. MDIS and CE agree within 0.20–0.24% at all levels, while the single shift diverges by up to 28%, 63% and 98% (90% quantile) at 10^{-3}, 10^{-4} and 10^{-5}. Relative variances: MDIS 4.3–6.1, CE 4.4–6.0, single shift 357–1 430. MDIS is built in 3–4 s against 43–55 s for CE, and CE costs 1 750–1 830 times more samples at $\varepsilon = 0.1$ and 18–19 times more at $\varepsilon = 0.01$.

7.4 Hubs

On 66 hubs (all of Northern Vietnam, 40 sampled from Guangdong), the hub estimate lies between $\max_r p_r$ and $\sum_r p_r$ in every case and agrees with 2×10^6-draw crude Monte Carlo on the 19 smallest hubs ($|z| \leq 1.67$). Ignoring the shared factors — treating routes as independent — overstates hub risk by a factor 1.75–2.2 in median and up to 3.2, and 27–37% of failing days have two or more

failing routes: standby capacity should be planned for clusters of failures. [Capacity buffers for $\gamma = 10^{-3}$ and 10^{-5} and region totals from the calibrated rerun.]

7.5 A note on quantum amplitude estimation

Quantum amplitude estimation offers a quadratic reduction in the number of simulator calls for estimating a probability [Brassard et al., 2002, Montanaro, 2015]. On the same 120 routes, with explicit constants for a rigorous schedule, it needs 0.003–0.5 times the calls of crude Monte Carlo, but 2.5–380 times more calls than classical IS once IS is available, and with an IS-equipped oracle it is cheaper than classical IS only in an idealised accounting at $\varepsilon \le 3\%$. On this problem the structure exploited by the mixture, not the quadratic speed-up, is what makes certification cheap.

8 Conclusion

Rare overflow of a pickup-and-delivery route has a simple anatomy — two end modes with closed-form scales, near-end stages close behind — and this anatomy decides when common IS shortcuts fail and how to build an estimator that needs no tuning. The resulting certificates are accurate to a fraction of a percent at 10^{-3}–10^{-6}, extend to route days without exact solution and to hubs with shared demand factors, and are cheap enough to run on regional networks. Open points: full proofs of the efficiency results at the stated generality, calibration of the capacity search, region-wide factors that couple hubs, and data-driven demand and travel models.

A Proofs

Proof of Proposition 1. $L_0 = D_0$ and $L_L = P_L$ are stage loads, and $D_k \le D_0$, $P_k \le P_L$ for all k. □

Lemma A1 (quantile). For the gamma law, $x(z) = G^{-1}(\Phi(z)) = \theta z^2/2 + \theta(2\kappa - 1)\log z + O(1)$ as $z \to \infty$, and $0 \le x(z) \le x(0)$ for $z \le 0$.

Proof sketch of Theorem 2. Upper bound: by Lemma A1, $D_k \le \frac{1}{2} Z^\top \Theta Z + R$ with R a sum of logarithmic terms; for $t < 1/\lambda$, $\mathbb{E}\exp(t\,\frac{1}{2} Z^\top \Theta Z) = \det(I - t\Theta^{1/2} C \Theta^{1/2})^{-1/2} < \infty$, so Chernoff's bound gives the rate $1/\lambda$, and R is negligible. The delivery and pickup blocks have disjoint inputs, so their moment generating functions multiply and L_k has rate $1/\lambda_k$. Lower bound: a ball of radius $O(1)$ around $(1+\eta)\sqrt{2c/\lambda}\,u^\star$, with $u^\star = B^\top \Theta^{1/2} v/\sqrt{\lambda}$ ($Z = Bu$, $BB^\top = C$, v the Perron vector), lies in the event. For F use Proposition 1. □

Proposition 4. *For a fixed gap* $\lambda_k < \lambda^\star$, $\mathbb{P}(L_k > c)/\mathbb{P}(F > c) = \exp(-c(1/\lambda_k - 1/\lambda^\star)(1 + o(1)))$.

Lemma A2. Let $h(u) = \frac{1}{2} u^\top M u$ with $M = A^\top \Theta A$, $A \ge 0$, eigenvalues $\lambda_1 > \lambda_2$, $\kappa = \lambda_1/\lambda_2 \ge 4$, $h_+(u) = \frac{1}{2}|\Theta^{1/2}(Au)_+|^2$, and $\tau = r v_1$ with $r^2 = 2c/\lambda_1$. Then $h_+(u) \ge c$ implies $|u + \tau|^2 \ge 4|\tau|^2$.
Proof. Write $u = a v_1 + u_\perp$, $b = |u_\perp|$. If $a \ge 0$, then $c \le \frac{1}{2}(\lambda_1 a^2 + \lambda_2 b^2)$, so $b^2 \ge \kappa(r^2 - a^2)$ and $|u + \tau|^2 \ge (1 - \kappa)a^2 + 2ar + (1 + \kappa)r^2$, a concave function of a whose minimum on $[0, r]$ is $\min((1+\kappa)r^2, 4r^2) = 4r^2$ for $\kappa \ge 3$; for $a \ge r$, $|u + \tau|^2 \ge (a + r)^2 \ge 4r^2$. If $a < 0$, then $Av_1 \ge 0$ gives $(Au)_+ \le (Au_\perp)_+$, so $b^2 \ge \kappa r^2 \ge 4r^2$. □

Proof of Theorem 3(a). $\tau^\star$ acts on the dominant block's inputs only and the other end event depends on the other block's inputs only, so $m_2 \geq \mathbb{E}[w]\,\mathbb{P}(\text{other end}) = e^{|\tau^\star|^2}\mathbb{P}(\text{other end}) = \exp(2c/\lambda^\star - c/\lambda_{\text{other}} + o(c))$, while $2\log\mathbb{P}(F > c) = -2c/\lambda^\star(1 + o(1))$. $\square$(b) follows from Proposition 1 and Lemma A2 applied to the block-diagonal form of the two end blocks; the logarithmic remainders are handled by Cauchy–Schwarz (sketch).

Theorem 5 (sketch)**.** *With a constant defensive weight, SMIS is logarithmically efficient if every retained block has $\kappa \geq 4$ and in every retained stage one block dominates the other by a factor of at least 4; stages with $\lambda_k \leq \lambda^\star/2$ need not be retained.*

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